Mr. Cooper

Interview Experience: Software Development Role

Package: 7.5 LPA (100000 bonus after completion of 1st Year, 150000 bonus after completion of 2nd year)

Round 1: Short Programming and MCQs

Duration: 3 hours
This round consisted of 30 questions, including 27 multiple-choice questions (MCQs) and 3 programming problems.

- MCQs: The questions were from key topics such as Data Structures, DBMS, Object-Oriented Programming (OOPs), Java, and Aptitude. Each question carried either 2, 4, or 6 marks.
- Programming Problems:
    - Two problem were asked from medium category(Rotten Oranges(matrix), Message Passing(Strings)
    - The third problem was hard Matrix problem.

Round 2: Long Programming Round

Duration: 2.5 hours

In this round, the task was to design and develop a **Bank Loan Management System** Application based on a provided Use Case. The application included five modules:

1. User Management
2. Application Approval
3. Repayment Process
4. Credit scoring & Risk Assessment
5. Reporting

The round was divided into several stages:

1. Data Model Design (45 mins)
    - The candidate had the choice to design the data model using techniques like ER diagrams or Class diagrams.
    - The Class Diagram approach was chosen.

2. System Architecture (15 mins)
    - Designed the system architecture, including the technology stack and data flow diagram.
3. OOP-Based Implementation (1 hour 15 mins)
    - Implemented the Class Diagram using Java.
    - The candidate was asked to code any two modules:
        - Application Approval
        - Repayment Process
4. Code Structure and Relationships (Evaluation Criteria)
    - The evaluators focused on OOP principles, class structuring, and the relationships between different entities.
    - Strong knowledge of Encapsulation, Inheritance, Polymorphism, and Association was essential.
5. SQL Query Writing (15 mins)
    - Based on the provided tables from the Use Case, SQL queries were written to manage data effectively.

Round 3: Technical HR – 1

Duration: 30 minutes

This round started with questions about my projects, specifically my **MERN stack music streaming project**. The interviewers asked:

- The **Node.js and React versions** I used.
- What **bundler** I used.
- How I implemented **music streaming** in the project.

Next, they moved on to problem-solving questions, including:

- **Two Sum** problem (Array-based problem)
- **Finding the lowest leaf node** in a Tree
- **Binary Search** implementation

They also tested my fundamental knowledge in **Operating Systems (OS), Database Management Systems (DBMS), and Computer Networking** with questions like:

- **OSI model and its layers**
- **Mutex and its role in concurrency**

Lastly, they asked about **OOPS concepts**, and I had to explain them using a **Chess Game** analogy.

Round 4: Technical HR – 2

Duration: 45 minutes

This round began with a **Use Case Analysis** related to a long programming round. The interviewers evaluated how I established relationships between the:

- **Loan Application Approval**
- **Repayment process**

Then, I was given two coding problems and had to **write the code** for them:

1. **Implement a stack using two queues** (only allowed to use `push()`, `pop()`, and `peek()` functions from the queue).
2. **Check if a singly linked list is a palindrome.**

After completing the coding questions, they presented a **SQL table** and asked me to **normalize it** by identifying redundancy and structuring it correctly.

Round 5: Personal HR

Duration: 30 minutes

This round was focused on my resume and projects. The discussion points included:

- Explanation of my **Android projects, open-source libraries, and GitHub profile**.
- A **live demo of one of my projects (Wiki BIT)**.
- Discussion on the **recent technologies I am interested in**.

They also asked **personal questions**, such as:

- **Family background**
- **Education details**
- **Hobbies**
- **Preferred work location**